Supplemental Material for
"Reading a magnet's Hamiltonian term by term: two-dimensional terahertz spectroscopy as vertex spectroscopy of $TmFeO_3$"

August 10, 2026

Contents

S1 The SU(3) qutrit and why its classical dynamics is quantum-exact

Tm^{3+} is $4f^{12}$, ${}^3H_6$ ($J = 6$). In the $C_s(m)$ crystal field of the $Pbnm$ $4c$ site the 13-fold multiplet splits into 13 singlets; the model keeps the three lowest, $|1\rangle, |2\rangle, |3\rangle$, with splittings $E_{12} = 2.068\,\mathrm{meV}$ ($0.50\,\mathrm{THz}$) and $E_{13} = 4.963\,\mathrm{meV}$ ($1.20\,\mathrm{THz}$), consistent with the measured lines $E_{12} + E_{23} \approx E_{13}$. The site state is the Gell-Mann vector $\lambda^a = \mathrm{Tr}(\rho\Lambda^a)$, $a = 1 \ldots 8$, and every single-ion term (CEF energies, external drives, exchange mean fields) is *linear* in the Λ^a. For a Hamiltonian linear in the generators the von Neumann equation $\dot\rho = -i[H, \rho]$ closes on $\vec\lambda$:

$$\dot\lambda^a \;=\; \Omega_{ab}(t)\,\lambda^b - \gamma_a\big(\lambda^a - \lambda^a_{\mathrm{eq}}\big), \qquad \Omega_{ab} = \tfrac{1}{2}\,\mathrm{Tr}\Big(\Lambda^a\,(-i)[H(t), \Lambda^b]\Big)\,. \tag{1}$$

The classical SU(3) propagation is therefore the *exact* quantum dynamics of the driven, damped three-level ion, and a Boltzmann-weighted ensemble of trajectories launched from the level seeds $|1\rangle, |2\rangle, |3\rangle$ is the exact thermal density matrix for single-ion drives. (This was verified explicitly against an independent density-matrix integrator; the two censuses agree to all digits quoted.) Damping is Bloch-type on the Gell-Mann pairs of each transition, $\gamma(\lambda^{1,2}), \gamma(\lambda^{4,5}), \gamma(\lambda^{6,7})$, with population relaxation $\gamma(\lambda^{3,8})$ toward the equilibrium populations captured at initialization.

The magnetically bright coordinate of the E_{12} block follows from the projected moment $PJ_zP = 5.264\,\lambda^2$: only λ^2 radiates magnetically, and the onsite splitting drives the planar precession $\dot\lambda^1 = -\Delta_{12}\lambda^2$, $\dot\lambda^2 = +\Delta_{12}\lambda^1$, so the E_{12} sector is a single driven oscillator

$$\lambda^2(t) = \int_0^t \sin[\omega_{12}(t - t')]\, h^1(t')\, dt', \tag{2}$$

where h^1 is the effective force a conversion vertex exerts on λ^1. All delay-axis structure of a cross peak lives in the τ-dependence of h^1.

S2 Crystallography and the projected Fe–Tm coupling

$\mathrm{TmFeO_3}$ is $Pbnm$ ($Z = 4$): Fe^{3+} at $4b$ (site symmetry $\bar{1}$; every Fe sits on an inversion center), Tm^{3+} at $4c$ (site symmetry the single mirror m_z). Inversion maps each Fe sublattice to itself and exchanges Tm sublattices in pairs; the local mirror is the physical $4c$ site mirror and generates the relative λ^5, λ^7 signs between Tm sublattices. The Fe order is G-type ($T_N \approx 632\,\mathrm{K}$) with DM canting; below the spin reorientation ($T_2 \approx 85\,\mathrm{K}$) the ground state is $\Gamma_2 = (F_x, C_y, G_z)$.

The Fe–Tm sector is built from the physical operator route

$$\boldsymbol{S},\ \boldsymbol{J} \;\longrightarrow\; H_{\mathrm{Fe-Tm}} = \boldsymbol{S}^T A\,\boldsymbol{J} \;\longrightarrow\; P\boldsymbol{J}P \;\longrightarrow\; \text{couplings to } \lambda^a, \tag{3}$$

never by assuming that a space-group rotation acts as a closed onsite SU(3) rotation in one fixed qutrit basis. Working in a *transported gauge* (qutrit bases on the four Tm sublattices related by the group operations themselves) makes every bond formula explicit and turns the irrep (Γ_1) projection into simple four-site sign sums. Time-reversal splits the Gell-Mann set into $\lambda^- = \{\lambda^2, \lambda^5, \lambda^7\}$ (odd) and $\lambda^+ = \{\lambda^1, \lambda^3, \lambda^4, \lambda^6, \lambda^8\}$ (even); every coupling term must be $\mathcal{T}$-even overall, which fixes how many Fe-spin or B-field legs may attach to each sector. The candidate couplings through cubic order in the fields are

family	operator form	sector	legs
K^-	$S_\alpha \lambda^a$	λ^-	1 Fe
ESJ	$E_\eta S_\alpha \lambda^a$	λ^-	1 E-photon + 1 Fe
κ^B	$B_\eta S_\alpha \lambda^a$	λ^+	1 B-photon + 1 Fe
W	$S_\alpha S_\beta \lambda^a$	λ^+	2 Fe

The full orbit formulas (all eight coefficients per bond, all four Tm sublattices, both time-reversal sectors) and the implementation prescriptions for global-axis versus local-frame storage are given in the long-form working note that accompanies the data (`tmfeo3_foundation.tex`, Secs. 3–14).

S3 The sharp requirement and the vertex scorecard

Two collinear pulses give $M_{\mathrm{NL}} = M_{AB} - M_A - M_B$; order by order only the mixed terms survive, so M_{NL} starts at second order and the leading robust magnetic-dipole contribution is third order. Combining Eq. (2) with the conservation rules at each vertex (energy; $\mathcal{T}$-parity; Γ_1) yields six requirements for a coupling to produce the geometry-A cross peak at $(q_{\mathrm{AFM}}, E_{12})$:

R1 (detection) reaches λ^1/λ^2 (the E_{12} block);

R2 (drive) is sourced by the pumped δS_y magnon;

R3 (parity) is $\mathcal{T}$-even (fixes the $\lambda^\pm$ sector);

R4 (irrep) is Γ_1-allowed in the Γ_2 state;

R5 (energy) carries the $0.90 \to 0.50$ THz mismatch (needs ≥ 2 non-λ legs);

R6 (M_{NL}) involves both pulses.

Table S1: Vertex scorecard. Only W and κ^B pass all six requirements; the final model uses the static W family alone, the geometry-B transfer being fed by the electric quadrature of the drive rather than by a field-assisted vertex.

vertex	legs	sector	R1	R2–R6	verdict
K^-	$1\mathrm{Fe} + 1\mathrm{Tm}^-$	λ^-	×	×	hybridizes only; Γ_1-blocked; reorients Γ_2
ESJ	$E + \mathrm{Fe} + \mathrm{Tm}^-$	λ^-	×	✓	real, but feeds E_{13}/E_{23}, not E_{12}
W	$2\mathrm{Fe} + \mathrm{Tm}^+$	λ^+	✓	✓	**winner**: rank-1, factorized peak
κ^B	$B + \mathrm{Fe} + \mathrm{Tm}^+$	λ^+	✓	✓	passes, but superseded (not in the final model)

K^- fails threefold. (i) Linear in S, it only rotates the harmonic normal-mode basis: coupled harmonic oscillators give diagonal peaks only; a static bilinear conserves energy and cannot convert 0.90 into 0.50 THz. (ii) The Γ_1 orbit sums vanish for every channel except K^-_{2y}: a static one-at-a-time scan at 0.2 meV leaves the energy, the Fe order, and the uniform $(\lambda^2, \lambda^5, \lambda^7)$ *bit-identical* to baseline for all components except $2y$. (iii) The one allowed channel K^-_{2y} is a transverse molecular field on the order parameter and reorients $\Gamma_2 \to G_y$ between 0.085 and 0.090 meV—it is not a perturbative vertex. Full-dynamics 2DCS on the inert channels (K^-_{2x} up to 1.5 meV, K^-_{2z} up to 2.0 meV) leaves the induced λ^2 at machine precision ($\leq 7 \times 10^{-15}$).

The two winners. The two-magnon vertex rectifies the pulse-A/pulse-B cross term,

$$h^1_{\mathrm{NL}}(t,\tau) = W_1^{yy} A^2 \left[\cos(q_{\mathrm{AFM}}\tau) - \cos(q_{\mathrm{AFM}}(2t+\tau))\right]\theta(t) \;\Rightarrow\; M^W_{\mathrm{NL}} \propto \cos(q_{\mathrm{AFM}}\tau)\left[1-\cos\omega_{12}t\right], \quad (4)$$

a clean factorized peak with no diagonal partner. The field-gated vertex kicks at the probe, $h^1 \propto B_y(t)S_y(t)$ with $S_y(0) = A\sin q_{\mathrm{AFM}}\tau$, giving $M^{\kappa}_{\mathrm{NL}} \propto \sin(q_{\mathrm{AFM}}\tau)[1-\cos\omega_{12}t]$: also on target. It is, however, not required. The one physical geometry-B pulse carries an electric quadrature, $E \parallel a$ acting through $d_x\lambda^1$, which writes the E_{12} coherence directly; the static W_1^{xz} at its master value then converts the stored coherence into the magnon, and the measured transfer peak is reproduced with $\kappa^B = 0$. The polarisation ownership of d_x supplies the geometry selection the B_z gate supplied before: the component does not exist in geometry A.

Exhaustion of the quadratic families. All remaining symmetry-allowed quadratic couplings were switched on one at a time in the full nonlinear dynamics and are exactly inert (differences at the 10^{-12} level or below): $W_4^{yy,xy,yz,xx,zz,xz}$, W_1^{xy}, $W_6^{yy,xz}$, κ^E_{2y}, κ^E_{5x}. Among the W_1 components: W_1^{yz} couples the qutrit to the A_y/C_y exchange modes (wrong magnon irrep—verified by the upper-branch ghost row at 1.4 THz under strong coupling), W_1^{zz} negligible, W_1^{xx} alive but floods the spectrum with an $\omega_t = 0.16$ THz comb, and W_1^{xz} is the on-target converter. No static component converts a *magnetically* written coherence without entering the mixer regime (level repulsion dragging the E_{12} line, excluded by 1D absorption at every strength—verified with the bare splitting recalibrated at each coupling so the dressed line stays at its measured position, which preserves $E_{12}+E_{23}=E_{13}$ automatically). The *electrically* written coherence requires no such regime: the master $W_1^{xz} = 0.01$ suffices, and the field-assisted family is retired by parsimony rather than by symmetry.

S4 Numerical protocol

Dynamics. Classical LLG for Fe (unit spins, $S = 5/2$ scale absorbed), SU(3) Bloch dynamics for the four Tm sublattices; RK4 with $\Delta t = 0.02$ code units; $1\times1\times1$ cell of the *Pbnm* magnetic unit (8 Fe + 4 Tm). Time is measured in $\hbar$/meV; a frequency f in THz corresponds to $E = f \times 4.136$ meV.

2DCS protocol. Delay scan $\tau \in [-120, 0]$ code units, $\Delta\tau = 0.1$ (0.2 for scans); detection window to $t_{\max} = 160$ for 0.01 THz resolution on ω_t. The nonlinear signal is the honest three-run subtraction $M_{\mathrm{NL}} = M_{AB} - M_A - M_B$ per delay; no pulse-window chunking; no reuse of the reference run across delays (both shortcuts were found to contaminate spectra and are documented pitfalls). The pump enters as a tabulated waveform (the measured field, times converted $t_{\mathrm{code}} = t_{\mathrm{ps}}/0.6582$, auto-centered and normalized).

Spectra and blind peak census. 2D spectra are $|\mathrm{FFT}_{\tau,t}|$ of M_{NL} with a detection window $t \geq 3$ (excludes pulse overlap). Cross-polarised channels use Hann windows on both axes; the same-polarised channel uses its documented convention, a cosine gate over the pulse-overlap region of the delay axis ($|\tau| < 6$ code units) with Gaussian t-apodization to 0.03. Spectra are lightly smoothed (1×0.5 pixels) and displayed on a linear amplitude scale. The delay axis is plotted physically ($\omega_T = -\omega_\tau^{\mathrm{FFT}}$; $+ =$non-rephasing). Peaks are accepted only by a blind census: 2D local maxima (maximum filter, 0.10 THz neighborhood) with prominence ≥ 1.5 over the local median background (0.34 THz annulus) and amplitude $\geq 5\%$ of the map maximum, merged within 0.09 THz. Rectified ($\omega_t \approx 0$) content is analyzed separately via $S_{\mathrm{DC}}(\tau) = \langle M_{\mathrm{NL}}\rangle_t$ with cubic drift removal. The $\omega_\tau = 0$ pump–probe row is retained in the displayed geometry-A panels, as in the measured maps, auto-scaled to the measured relative row amplitude (the raw row is dominated by broadband pump–probe drift; the applied factor is recorded in the census file); rows with $|\omega_\tau| < 0.18\,\mathrm{THz}$ are excluded from all censuses, since they carry pump–probe rather than delay-coherence information. Radiation bookkeeping used throughout: for $k \parallel b$, M1 dipoles radiate $E \propto \hat{k} \times m$, so the $E \parallel c$-detected channel carries $d_z(\lambda^2) + m_x$ and the $E \parallel a$-detected channel carries $d_x(\lambda^{5,7}) + m_z$.

Thermal ensembles. Boltzmann mixtures of trajectories from the level seeds $|1\rangle, |2\rangle, |3\rangle$ (Sec. S1). Two documented pitfalls: (i) any pre-dynamics energy minimization collapses excited-level seeds and must be disabled; (ii) population damping $\gamma(\lambda^{3,8})$ relaxes toward the equilibrium captured at initialization, which must therefore be captured per-seed.

S5 Experimental inputs

Table S2: Measured mode parameters and the derived simulation inputs. CEF damping is Bloch on the Gell-Mann pair of each transition, $\gamma_a[\mathrm{meV}] = \gamma[\mathrm{THz}] \times h$; magnon damping is Gilbert, $\alpha = \gamma/2\nu$ fixed by the q_{AFM} line. Emission weight is $\sqrt{f}$ (E1) for the CEF lines and M1 for the magnons. The last column is the measured pulse spectral amplitude at each mode.

mode	ν [THz]	f	$\sqrt{f}$	γ [THz]	damping (sim)	pulse amp.
q_{FM}	0.38	0.08	0.28	0.005	$\alpha = 1.7 \times 10^{-3}$	0.92
q_{AFM}	0.90	7×10^{-4}	0.026	0.003	$\alpha = 1.7 \times 10^{-3}$	0.68
E_{12}	0.50	6.5	2.55	0.038	$\gamma(\lambda^{1,2}) = 0.157\,\mathrm{meV}$	1.00
E_{23}	0.70	6.6	2.57	0.10	$\gamma(\lambda^{6,7}) = 0.414\,\mathrm{meV}$	0.85
E_{13}	1.20	8.7	2.95	0.05	$\gamma(\lambda^{4,5}) = 0.207\,\mathrm{meV}$	0.36

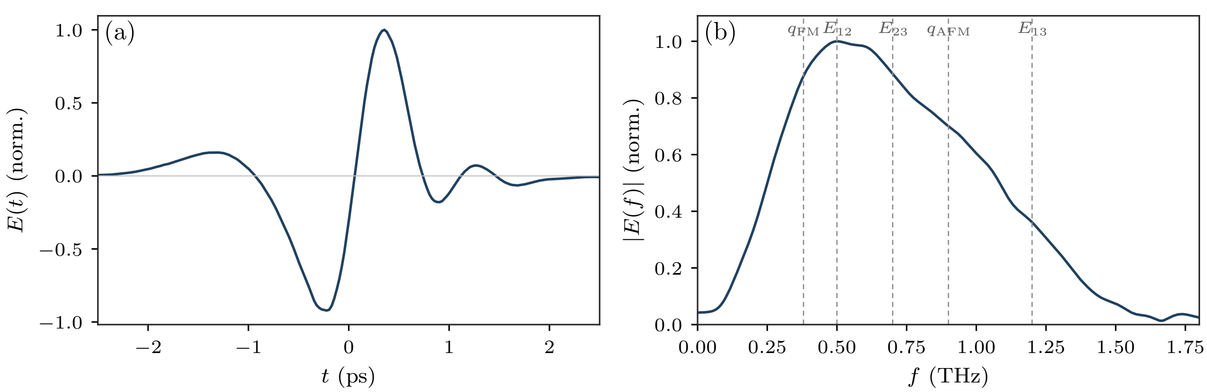

Figure S1: The measured THz pump. Left: single bipolar cycle ($\sim 1.2\,\mathrm{ps}$, no DC). Right: amplitude spectrum peaked at $0.53\,\mathrm{THz} \approx E_{12}$, covering all five modes. The absence of DC content removes the quasistatic-tilt legs that a Gaussian stand-in pulse would add; the measured waveform is the cleanest configuration.

Two consequences organize the readout physics. First, excitation scales as E^2 and detection as E^1 (third order). Second, the CEF lines are E1-bright while q_{AFM} is E1-dark but M1-bright: a magnon is therefore read out either by borrowing a CEF electric dipole through an Fe–Tm vertex (geometry A) or through its own magnetic dipole at the magnon frequency (geometry B). Using the magnetic g-factor instead of $\sqrt{f}$ for the magnon electric emission over-weights it by $\sim 100\times$ and buries the CEF peaks under the magnon diagonal; the oscillator-strength weighting resolves this without any tuning.

S6 Displacive verification of the W mechanism

The rectified two-magnon force is a suddenly switched step: its edge carries spectral weight $\propto e^{-\omega_{12}^2\sigma^2/2}$ at the CEF frequency, i.e. the E_{12} quantum is drawn from the pulse bandwidth (intra-pulse difference-frequency/impulsive-Raman), while the magnon pair carries only the phase memory $\cos q_{\mathrm{AFM}}\tau$. Verified directly: stretching the pulse $\sigma = 0.25 \to 1.0$ at fixed amplitude collapses the peak by 4.5 decades ($51.7 \to 1.7\times10^{-3}$), while switching the direct Tm drive on/off changes it by $< 10^{-4}$ (pure-W origin). The same sweep resolves the second Fe leg: impulsive pulses use the resonant probe magnon (peak $\propto A^2$); longer pulses cross over to the field-following quasistatic tilt (peak $\propto A^1$). The measured no-DC waveform suppresses the quasistatic legs, which is why the experimental pulse yields the cleanest spectra.

S7 Geometry B: the transfer and the post-drive buildup

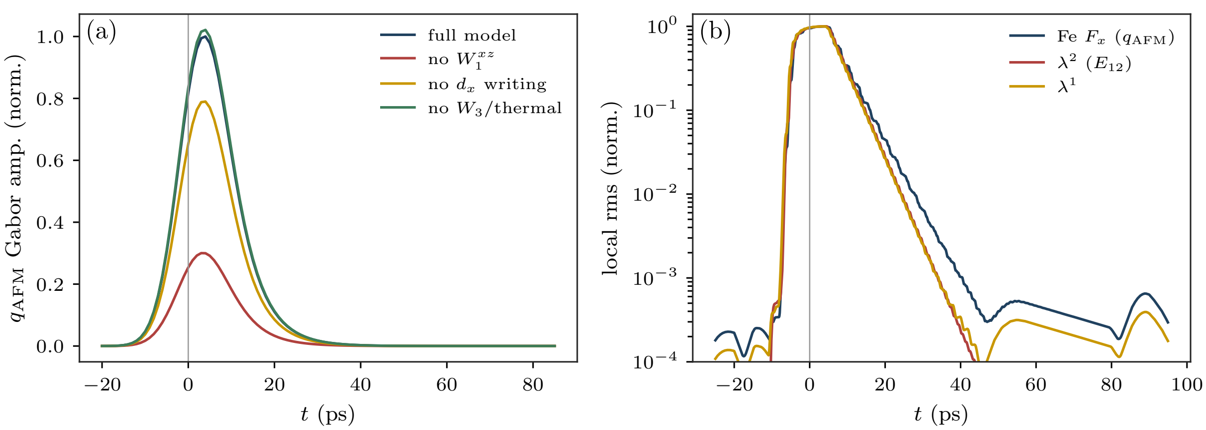

Figure S2: No coherence-fed post-pulse buildup once the measured linewidths are enforced (single measured pulse, geometry B; Gabor amplitude at 0.90 THz, $\sigma = 5$ ps). (a) The q_{AFM} amplitude peaks *at* pulse end and decays with $\tau_{\mathrm{env}} \approx 8\,\mathrm{ps} = T_2(E_{12})$ in the full model and in every term ablation; the terms set the absolute scale (removing W_1^{xz} or the d_x writing lowers the peak, removing W_3/thermal changes nothing) but none produces post-pulse growth. (b) All coherent reservoirs die within their measured T_2 (log scale); the weak late-time floor is the incoherent thermal tail, orders of magnitude below the impulsive peak.

The geometry-B cross channel is analysed on the Fe m_x dipole with its Tm part identically zero (subalgebra closure), so it reads the magnon amplitude directly. Its transfer peak $(E_{12}, q_{\mathrm{AFM}})$ is the energy face read backward: a stored $1 \leftrightarrow 2$ coherence handed to the quasi-antiferromagnetic magnon. Isolated-drive runs settle the anatomy. With the coherence written *magnetically* ($\mu_z\lambda^2$ only) at the master $W = 0.01$ the conversion is inefficient and the map fills with drive-squared diagonals; adding the *electric* quadrature of the same physical pulse ($d_x\lambda^1$, the component only $E \parallel a$ possesses) produces the transfer row at the *unshifted* E_{12} position, at the measured amplitude for $w_{E1} = 0.54$, with strays below 0.17. The interference of the two quadratures places the row at $\omega_\tau = 0.52$ for $\mathrm{sign}(d_x\mu_z) < 0$, as measured. The leave-one-out runs (Sec. S14) quantify the shares at the operating point: removing the electric writing halves the transfer (0.49), removing W_1^{xz} costs a quarter (0.76), the remainder flowing through the magnetically written quadrature and the W_1^{yy} mixing route ($q_{\mathrm{FM}}+E_{12} = 0.88$, unresolved from q_{AFM}); in every ablation the emission stays pinned at 0.900 THz—all routes ultimately ring the real q_{AFM} mode. No field-assisted vertex appears anywhere

($\kappa^B = 0$; its inertness in these geometries is bitwise, Sec. S15).

The measured signal, however, continues to grow after the pulse, and within this coupling space no channel reproduces growth at the peak's own scale (Fig. S2): every coherent pathway inherits its source's lifetime, and with $T_2(E_{12}) = 8\,\mathrm{ps}$ everything coherent is over within $\sim 10\,\mathrm{ps}$. The one reservoir that outlives every T_2—the CEF populations, fed incoherently by the dissipated pulse energy and acting on the magnon through the population channel W_3—produces a persistent, drive-uncorrelated tail roughly four decades below the impulsive peak. The buildup at the peak's own scale therefore stands as the one timing observable outside this coupling space; the same incoherent reservoir is read optically by the rectified $(\pm E_{12}, \omega_t \to 0)$ line of the same-polarised channel, whose strength relative to (E_{12}, E_{23}) calibrates $c_{\mathrm{heat}} W_3$.

S8 Propagation: self-absorption of the electric-dipole light

The sample is optically thick at the strong $E1$ crystal-field lines ($f = 8.7$ at E_{13}, depth $\simeq 6$; depth $\simeq 1$ at E_{12}), and the same transmission filters the incoming field and the emitted $E1$ light; magnetic-dipole radiation passes freely. The filter is implemented as a Lorentzian optical depth on the $E1$ components,

$$T(\omega) = \exp\Big[-\frac{d}{1 + \big((\omega - \omega_{13})/\gamma_{13}\big)^2} \Big], \qquad d \simeq 6, \tag{5}$$

applied to the emission column *and* to the delay-axis free induction decay of any electrically written coherence. Three measured structures follow with no further freedom. (i) The two E_{13}-emission features sit at $\omega_t = 1.15$ and $1.25\,\mathrm{THz}$, one absorption linewidth below and above the 1.20 line: self-reversal wings of an optically thick line, whose centre is dark ($T \simeq 0.002$). (ii) The two-quantum (E_{13}, E_{23}) row, which the free-ion model robustly produces at 0.33 (the pump reaching ρ_{13} at second order), is removed *exactly* by the FID reabsorption on the delay axis—the model value falls to 0.00 with the filter, and the measurement contains no $\omega_\tau = E_{13}$ feature. (iii) The SFG tone of Sec. S10 escapes at 1.41, one linewidth *outside* the opaque window ($T = 0.68$), which is why it is visible while the line centre is not. Every wing amplitude scales with sample thickness: a thin sample revives the E_{13} centre and the two-quantum row *together*, the decisive propagation test.

S9 The same-polarised channel at the final operating point

The channel is analysed by the minimal $(E \parallel c, H \parallel a)$ operator

$$O_1 = F_x + 0.913\,\lambda^7 + 4.4\,\lambda^6 + g_d(\langle F_x \rangle + \delta F_x)\lambda^2, \tag{6}$$

with $\mu^x_{13} = 0$ and $d_z(1 \to 3) \leq 0.006$ (bounded by the weak 1.15 wing and dropped), the Fe weight bounded above by the absent magnon diagonal ($c_{\mathrm{Fe}} \to 0$; the diagonal sits at 0.13, below the digitisation threshold), and the propagation filter of Sec. S8 applied to the $E1$ components.

The fully dark $1 \to 3$ *dipole is measured, not assumed.* The measured map contains delay frequencies $\omega_\tau \in \{E_{12}, q_{\mathrm{AFM}}\}$ only; a bright μ^z_{13} would store an E_{13} coherence at first order and produce an (E_{13}, E_{23}) peak with exactly the dipole product of the observed (E_{12}, E_{23}). The analyser moreover physically contains $\mu_x \lambda^5$, and at the point-charge value $\mu^x_{13} = 2.39$ that single term would exceed the whole measured map by two orders of magnitude: μ^x_{13} fits to zero. The transition is magnetically dark in every polarisation; its large measured oscillator strength is electric and is precisely what makes the sample opaque at $1.20\,\mathrm{THz}$.

Final accounting. With the cross channels held fixed:

peak	experiment	model	mechanism
(E_{12}, E_{23})	1.00	1.00	coherence→population→ ρ_{23}, non-rephasing
$(E_{12}, 0.40)$	0.45	0.40	cascade tone $2(E_{23}-E_{12})$, adjacent to $q_{\rm FM} = 0.38$
$(E_{12}, 0.90)$	0.40	0.43	cascade tone $2E_{23}-E_{12}$, degenerate with $q_{\rm AFM}$
$(q_{\rm AFM}, E_{12})$	0.30	0.39	magnon→Tm via $d_z^{\rm eff}$ (reciprocity partner)
$(q_{\rm AFM}, q_{\rm AFM})$	absent	0.13	magnon diagonal; the null bounds $c_{\rm Fe}$
(E_{13}, E_{23}) 2Q	0	0.00	killed exactly by FID reabsorption
$(E_{12}, 1.15)$	0.15	0.02	E_{13} wing; bounds $d_z(1\to 3)$
$(q_{\rm AFM}, 1.41)$	0.30	0.31	SFG tone $q_{\rm AFM}+E_{12}$ (Sec. S10)
$(q_{\rm AFM}, 0.40)$	obs.	0.43	DFG tone $q_{\rm AFM}-E_{12}$
$(E_{12}, 1.41)$	obs.	0.36	E_{12}-delay partner of the SFG
$(-E_{12}, E_{23})$	< 0.15	~ 0.7	**open**: residual R1

Cascade tones at the magnon frequencies. The two E_{12}-delay features at 0.40 and 0.90 survive the *no-magnon* ablation at 0.89–0.97 and die without the λ^7 drive leg (0.05–0.08): they are intrinsic tones of the driven three-level cascade, $2(E_{23}-E_{12})$ and $2E_{23}-E_{12}$, numerically coincident with $q_{\rm FM}$ and $q_{\rm AFM}$. The genuine Tm→magnon transfer is the $\lesssim 10\%$ Zeeman-sensitive remainder. Positionally the low tone sits at 0.40, a full linewidth above $q_{\rm FM} = 0.38$—a direct experimental discriminator—and cascade tones survive the spin reorientation with CEF width, while magnon sidebands would soften and sharpen.

Exact single-ion verification. A standalone integrator of the 8-dimensional $\vec{\lambda}$ equation (same CEF energies, dampings, measured pulse, same census pipeline; ~ 13 s per 2D map) reproduces the lattice E_{12}-family census exactly, proving the family is single-ion physics and enabling exhaustive scans. Two facts emerge: the rephasing/non-rephasing ratio of the (E_{12}, E_{23}) transfer has a floor of 0.47–0.58 under this pulse across all drive strengths, dipole phases and dampings—the *rephasing remnant is an exact property of any three-level ion driven by this waveform*, and it is the one residual (R1) the model does not remove, the measured negative-delay side carrying only the rectified and $q_{\rm FM}$ features. The SFG/DFG tones contribute rephasing partners of their own there (model 0.5–0.7). This residual, the wing amplitudes, and the geometry-B $d_z^{\rm eff}$ doublet strength all funnel into one measurable object: the sample's polarisation-resolved optical depths at the three CEF lines.

S10 The dynamical $d_z^{\rm eff}$: internal magnon–CEF SFG and DFG

The order-induced dipole is the ESJ vertex $-g_d E_z(F_x\lambda^2)$; condensing its spin leg on $\langle F_x\rangle$ gives the drive and linear-emission avatars, and the retained fluctuation $g_d\,\delta F_x\lambda^2$ is a bilinear emitter radiating at $q_{\rm AFM} \pm E_{12} = 1.40/0.40$ THz with the magnon on the delay axis. Because the trajectory pack stores the full time evolution of every coordinate, any candidate product moment X_iX_j is evaluable post hoc with no new simulation: $P_{\rm NL} = X_i^{AB}X_j^{AB} - X_i^{B}X_j^{B} - X_i^{0}X_j^{0}$ per delay, followed by the standard census. A sweep of all 33 symmetry-classified Fe–Fe (F/G bilinears) and Fe–Tm ($F/G \times \lambda^{1,2}$) products gives a sharp verdict:

- Every single-coordinate (linear) channel and all four remaining on-site quadratic moments of the $4c$ site ($\lambda^1\lambda^6, \lambda^2\lambda^7$ as d_z; $\lambda^1\lambda^7, \lambda^2\lambda^6$ as m_x) radiate at $E_{12}+E_{23} = 1.20$ THz with an $(E_{12}, 1.20)$ partner 10–12× their $q_{\rm AFM}$-row feature—opposite to the measured ratio. The measured smallness of $(E_{12}, 1.20)$ bounds all four at once.
- The Fe–Fe striction products (G_xG_y, F_yG_y) land at $q_{\rm AFM}+q_{\rm FM} = 1.28$ with the right ratio but carry full-strength combination delay rows at $|\omega_\tau| = 1.05/1.49$ that the measurement lacks.

- $\delta F_x \lambda^2$ (with its m_x-type siblings) is the *only* channel of the catalogue whose 2D map is led by the q_{AFM}-delay row (partner ratio 1.4–1.6 instead of the universal ~ 0.1): both of its factors are written at first order by the same pulse, so the stored magnon pays no conversion penalty. The term-ablation runs confirm exactly this—the tones survive removal of both W components (0.95–1.02) and die only with the Zeeman or electric drive legs.

One number is fitted: the condensation-ratio correction $s = 0.42$ on the naive trajectory weight $\delta F_x/\langle F_x\rangle$ (the model's canting angle, $\langle F_x\rangle = 8.4 \times 10^{-3}$ per spin, overestimates the ratio by about two). The sum tone then lands at the measured $(q_{\mathrm{AFM}}, 1.41) = 0.30$; the difference tone at 0.43 merges into the measured $(q_{\mathrm{AFM}}, E_{12})$ blob; the E_{12}-delay partner appears at 0.36. Together with the on-site $\beta\lambda^1\lambda^2$ these complete the $\chi^{(2)}$-like triad of the inversion-free rare-earth site in emission—SHG (degenerate), SFG and DFG (non-degenerate, cross-subsystem)—with the mixing element a bilinear term of the material's transition moment, not a crystal-optical $\chi^{(2)}$ acting on light. The triplet is over-constrained: locked ratios, position at $q_{\mathrm{AFM}}+E_{12}$ exactly (at 1.20 it would be a quadratic moment, and the dark E_{13} window separates the cases), tracking of the softening magnon, and collective collapse at the reorientation with everything that rides $g_d \propto \langle F_x\rangle$.

S11 One drive for one experiment: the operating point

The two geometry-A channels are the same measurement—same $H \parallel a$, $E \parallel c$ excitation, differing only in the detected polarisation—so they share one drive, and the Tm and Fe drive amplitudes are tied by the g-factor ratio: $A_{\mathrm{Tm}} = g_{\mathrm{ratio}}|v|A_{\mathrm{Fe}}$. Fluence is therefore the single drive parameter, and the cross channel measures it. At strong drive ($A_{\mathrm{Fe}} = 1.2$) the channel is nonperturbative: the E_{12} diagonal reaches 1.45 of the $(q_{\mathrm{AFM}}, E_{12})$ peak and a forest of multi-magnon strays appears at $\omega_\tau = 0.63, 0.72, 0.99, 1.25, 1.51$ THz, none observed. Lowering the fluence removes them monotonically (the diagonal falls to 0.56, 0.16, 0.06 at $A_{\mathrm{Fe}} = 0.4$, 0.12, 0.04), so the experiment sits in the perturbative regime; the production operating point is $A_{\mathrm{Fe}} = 0.12$ (geometry B independently at 0.10). The former second drive dial—a residual μ_{13}^z admixture—is gone: the $1 \to 3$ transition is fully dark as a *measured* statement (Sec. S9), and the same-polarised hierarchy that the admixture once tuned is fixed instead by the propagation filter.

The second cross peak is second-harmonic emission

The second geometry-A cross feature at $(E_{12}, 2E_{12}) = 0.96$ is the quadratic-moment harmonic: $m_z = \mu\lambda^2 + \beta\lambda^1\lambda^2$, allowed because the Tm $4c$ site lacks inversion (Fe at $4b$ is excluded). Evaluated on the trajectories the quadratic channel contributes essentially one peak, $5.8\times$ its own (E_{12}, E_{12}) diagonal, identical in local and global sublattice frames. Under term ablation it survives removal of the Zeeman drive and of both W components at 1.00–1.03 and dies to 0.005 without the electric E_{12} writing: pure stored-coherence SHG. β is fixed once from the observed parity $(E_{12}, 2E_{12}) \simeq (q_{\mathrm{AFM}}, E_{12})$ and re-used everywhere, including the zero-parameter geometry-B same-polarised prediction.

S12 Reciprocity, settled

Light couples to matter through one operator per polarisation state, and the same operator radiates. The final model satisfies this exactly at every level the data can test. Each analyser state is one operator serving excitation and detection alike:

$$O_2 \equiv (E \parallel a, H \parallel c) = S_z + 5.264\lambda^2 + 2.84\lambda^1 + \beta\lambda^1\lambda^2,$$
$$O_1 \equiv (E \parallel c, H \parallel a) = S_x + 0.913\lambda^7 + 4.4\lambda^6 + g_d(\langle F_x\rangle + \delta F_x)\lambda^2,$$

with the channel map A-cross $= O_1 \to O_2$, A-same $= O_1 \to O_1$, B-cross $= O_2 \to O_1$, B-same $= O_2 \to O_2$. Shared exactly between drive and readout: the operator content, the within-family dipole ratios, and the propagation filter. Completing the drive with the full electric components of its operator changes no pathway ratio—a direct scan gives exact nulls—so drive-side reciprocity is exact and free. Three cross-family scales are calibrated once and used consistently on both sides: $w_{E1} = 0.54$ (measured by the geometry-B transfer), g_d (drive avatar, linear-emission avatar, and the condensation ratio of Sec. S10), and c_{Fe} (bounded by the absent magnon diagonal).

S12.1 The $E1/M1$ balance: detection-blind, drive-measured

w_{E1} is the ratio of the electric to the magnetic amplitude of the $1 \leftrightarrow 2$ transition; both are allowed at the inversion-free $4c$ site. In *detection* the atlas cannot fix it, for a structural reason: λ^1 and λ^2 are the two quadratures of one coherence, their magnitude spectra agree to 8–11%, and adding one to the other rescales the resonant response without changing its shape. Sweeping the detection weight over a factor of 50, with β refitted at each step from the observed geometry-A parity:

w_{E1}/μ	0	0.19	1.0	1.9	5.0
β	31.5	38.1	66.3	97.6	205.5
A cross $(q_{\mathrm{AFM}}, E_{12})$	1.00	1.00	1.00	1.00	1.00
A cross $(E_{12}, 2E_{12})$	0.97	0.97	0.97	0.97	0.98
A cross $(-q_{\mathrm{AFM}}, E_{12})$	0.52	0.51	0.49	0.48	0.48
A cross (E_{12}, E_{12})	0.33	0.34	0.35	0.36	0.37

Every geometry-A cross amplitude moves by at most 0.04 across the whole range; β absorbs the change. What fixes w_{E1} is the *drive*: the same d_x matrix element that would radiate also writes, and the electric component of the geometry-B transfer is linear in it; the measured ratio returns $w_{E1} = 0.54$.

S13 The complete atlas and its censuses

Blind censuses at the final operating point, normalised per channel [amp, (ω_T, ω_t); $+ =$ non-rephasing]:

channel	census	reading
A cross ($O_1 \to O_2$)	1.00 $(+0.90, 0.49)$; 0.96 $(+0.49, 1.00)$; 0.50 $(-0.90, 0.49)$; 0.32 $(+0.49, 0.49)$; 0.28 $(+0.90, 0.89)$	W magnetoelectric anchor; $2E_{12}$ SHG; rephasing partner; E_{12} diagonal; magnon-forced sideband (both W components)
B cross ($O_2 \to O_1$)	1.00 $(+0.38, 0.90)$; 0.42 $(+0.52, 0.90)$; 0.15 $(+0.88, 0.90)$; 0.13 $(-0.38, 0.90)$	magnon–magnon; electric-write $\to$ W transfer; q_{AFM} diagonal (genuine Fe M1); rephasing partner
A same ($O_1 \to O_1$)	1.00 $(+0.49, 0.70)$; 0.74 $(-0.49, 0.69)$; 0.54 $(-0.49, 0.41)$; 0.43 $(+0.90, 0.40)$; 0.39 $(+0.90, 0.50)$; 0.36 $(+0.51, 1.41)$; 0.31 $(+0.90, 1.41)$	population-route anchor; R1 remnant; DFG rephasing partner; DFG tone; d_z^{eff} readout; SFG partner; SFG tone
B same ($O_2 \to O_2$)	1.00 $(+0.51, 1.00)$; all others ≤ 0.07	$2E_{12}$ SHG alone: the zero-parameter β prediction

S14 The necessity audit: runs and ratios

Every component was subjected to a leave-one-out test: drive legs and Hamiltonian vertices by re-running the solver with one term removed (single ground-state seed; feature ratios to baseline), detection terms and filters by dropping one composite term. Drive-leg ablations rescale the remaining vector so all other effective fields are unchanged.

removed	what dies (ratio to baseline)	verdict
Fe Zeeman leg	A-cross anchor 0.02; $(q_{\mathrm{AFM}}, E_{12})$ 0.03; SFG/DFG 0.03	required
d_z^{eff} drive	anchors and SHG 0.005; tones 0.08–0.13	required
$\mu_x\lambda^7$ drive	anchor 0.02; cascade tones 0.05–0.08	required
$W\,S_y^2\lambda^1$	A-cross anchor 0.03; forced sideband 0.43	required
$W\,S_xS_z\lambda^1$	sideband 0.64; B transfer 0.76	required (shared)
d_x drive (geom. B)	transfer 0.49	dominant leg
β (detect)	SHG 0.95 $\to$ 0.01	required
λ^6 (detect)	same-pol anchor 1.00 $\to$ 0.06	required
$\delta F_x\lambda^2$ (detect)	SFG/DFG triplet 0.02–0.05	required
$\langle F_x\rangle\lambda^2$ (detect)	$(q_{\mathrm{AFM}}, E_{12})$ blob re-centres at 0.40	required (position)
T_{13} column / delay-row	0.67 line-centre peak appears / 2Q row returns at 0.32	required
κ_{1y}^B	nothing: trajectories bitwise identical	inert (exact)
W_3 + thermal channel	all coherent features $\leq 5\%$	atlas-inert; incoherent only
λ^4 (detect)	only the 1.15 wing (0.03 $\to$ 0.02)	marginal; kept as bound
F_z (detect)	nothing ($F_z \equiv 0$ in A)	symmetry spectator
$\lambda^1 \leftrightarrow \lambda^2$, $\lambda^6 \leftrightarrow \lambda^7$ swaps	maps agree to 8–11%	quadrature-equivalent

The audit also fixed two assignments now standard in the main text: the same-polarised E_{12}-row features at 0.40/0.90 are driven-qutrit cascade tones (they survive the no-magnon run), and the A-cross magnon sideband and geometry-B transfer are shared between the two W components rather than owned by one.

S15 Code-level verification

The solver implementation was verified against the theory independently of the physics audit; the full record is in the long-form note. In brief: all nine Gell-Mann structure constants are implemented at their standard values and the qutrit equation of motion is the exact von Neumann equation (pump-only trajectories ring at 0.503/0.700/1.200 THz as required); the coupling parsers make the $\mathcal{T}$-classes structurally unrepresentable ($SS\lambda$ readers expose only $\lambda^{1,3,4,6,8}$, K^- only $\lambda^{2,5,7}$); the transported-gauge frames and inversion-partner signs match the crystallographic derivation; the trilinear $S^TWS\lambda$ is distributed as the exact gradient of one energy (the Fe site receives both spin legs, the Tm site S^TWS once), so forward and backward conversion follow from one term with no independent back-action parameter; the field-assisted vertices are gated by lab-frame field components, the mechanism of the bitwise κ^B inertness; and the staggered four-sublattice combination of the Tm coordinates is zero to the last floating-point digit, statically and at every delay, in both geometries—the four transported Tm ions are exactly equivalent, so sublattice projection signs can only renormalise whole channels, never ratios or positions. The stored reference trajectory is the pump-only M_0, making the extracted signal literally $M_{\mathrm{NL}} = M_{AB} - M_A - M_B$, and the equilibrated reference is the exact Γ_2 configuration.

S16 Data and code availability

Everything needed to reproduce every spectrum in this work, in both the time and the frequency domain, is collected in `paper/data/`.

The central object is `atlas_construction_pack.h5`, a single self-documented file (all conventions in its own HDF5 attributes) containing: the *time domain* $M_{\mathrm{NL}}(\tau, t)$ of every coordinate at full resolution, for both geometries and all three thermal seeds; the two *product* channels $(\lambda^1\lambda^2)_{\mathrm{NL}}$ and $(F_x\lambda^2)_{\mathrm{NL}}$, which are supplied explicitly because a product of nonlinear signals is not a function of the stored M_{NL} (it requires the two-pulse, probe-only and pump-only trajectories separately) and without which the SFG/DFG triplet cannot be reconstructed; the *frequency domain* $|\mathrm{FFT}_2|$ maps of the four published panels with their THz axes; and a *hybrid* group holding $|\mathrm{FFT}_\tau|$ against real detection time. The reader `make_atlas_from_pack.py` rebuilds all four panels and their blind censuses from that file alone—no solver, no run directories—and reproduces the published censuses to float32 storage precision. The same content is mirrored in conventional formats for collaborators who do not use HDF5: `matlab/` (per-run `.mat` time-domain files plus `spectra.mat` and `mixed_domain.mat`, each carrying an embedded `readme`) and `csv/` (gzipped CSV of every array with plain-text axes). Also included: `READOUT_CHANNELS.md` (the exact drive and detection operator of each channel, with dataset names), `OPERATORS.md` (the complete operator inventory: what is in the model, what is bounded, what is excluded and why), `params/` (the exact solver inputs of the five production runs), the published censuses and the digitised experimental map.

The long-form derivation note (`tmfeo3_foundation.tex`: transported-gauge orbit formulas, vertex diagrams, the complete written-out Hamiltonian, and the necessity and code-level audits in full) accompanies the data; every simulation uses the public `ClassicalSpin_Cpp` solver.

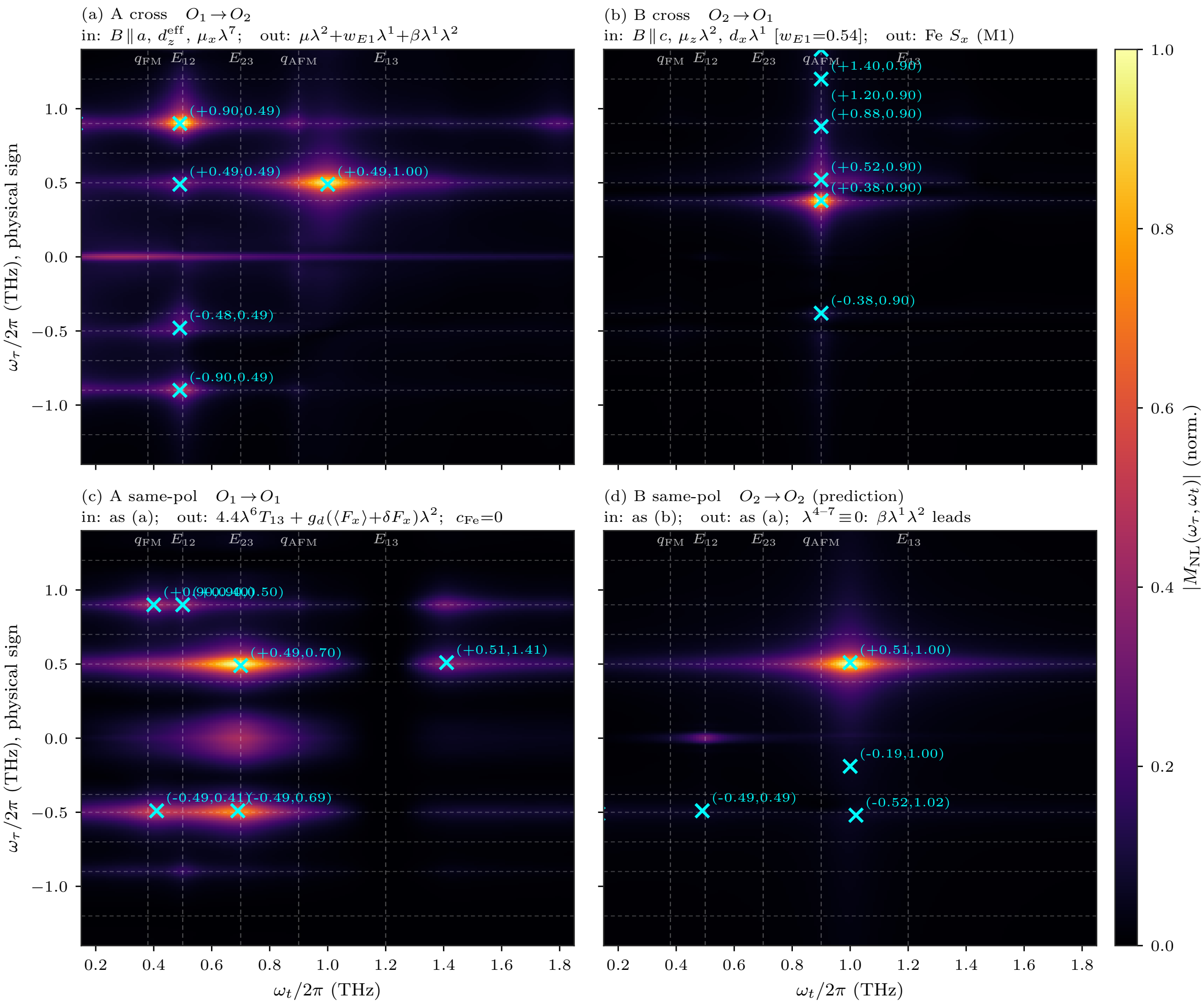

Figure S3: All four polarisation channels from the single consolidated Hamiltonian at one operating point (linear amplitude colour scale; blind-census crosses; the $\omega_\tau = 0$ pump–probe row retained in the geometry-A panels, auto-scaled, and excluded from the census). Each panel states its exact drive and detection. Numerical censuses in `data/census_atlas_kBfree.json`.